

Deployable Origami Solar Array (DOSA) For Space Colonization

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Artemis Mission Summary

The Solar Folds team provides an innovative solution to tackle key issues faced in space exploration through an origami-inspired solar array. Our lightweight system enhances portability as opposed to conventional solar arrays. The development of the deployable solar array perfectly aligns with the Artemis Mission initiatives. The system emphasizes space-exploration-safe deliverables with minimal stress points, compact stowage. Our unique starshade inspired design not only provides effective solar energy capture but also embodies the innovative spirit of space exploration.

Problem Statement

The Artemis Mission seeks to enhance lunar exploration, investigating potential living and working conditions on the moon in preparation for future human missions to our Lunar neighbor and beyond. Our system seeks to address the following criteria:

- Optimized mobility
- Retraction and redeployment as needed
- Lightweight design with enhanced scalability

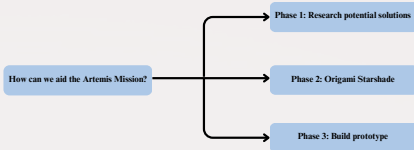
Objective

The Solar Folds' goals sought to address key Artemis Mission requirements for portable and efficient solar power production on the lunar landscape. Our deployable solar array structure is derived from origami engineering principles, allowing innovative and space-efficient solutions. The array maximizes surface area for energy capture while minimizing launch mass and stowage volume. Key design attributes include durability in lunar environments, compactness, and scalability. By integrating these features, our solar array provides a viable solution for sustainable power generation in future lunar exploration endeavors.

Method

This flowchart depicts our three-phase process to advance the Artemis Mission, focusing on innovative design and practical implementation.

Table 1: Engineering Design Chart



Prototype

Our starshade inspired origami solar array addresses critical issues such as space, weight, and deployment capabilities.

Table 2: Design Features

Feature	Benefit	Relevance to Artemis Mission
Compactness	Easily folds for transport	Save space on launch
Deployability	Expands for large coverage	Increase solar capture
Lightweight	Reduces mass	Lowers launch costs, adds payloads
Scalability	Customizable shapes and sizes	Tailored for desired needs
Innovation	Novel origami principles in design	Showcases cutting-edge technology



Figure 1.1: Top View Closed



Figure 1.2: Top view half open



Figure 1.3: Top view fully expanded

Results

Our shift to a more research-centric approach resulted in valuable insights from simulations, confirming the feasibility of the origami solar array under lunar conditions and identifying critical environmental factors.

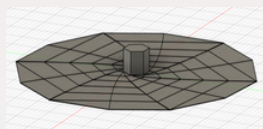


Figure 2.1: Geometric Model of Structure (Side View Under 90 Degrees)

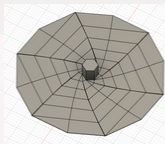


Figure 2.2: Geometric Model (Side View Above 90 Degrees)

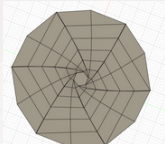


Figure 2.3: Geometric Model (Top View)

Next Developmental Phases:

- Physical prototyping to build tangible models
- Testing in simulated lunar environments to assess practical performance
- Empirical refinement to optimize design based on test results
- Stress and fatigue analysis to evaluate the long-term endurance and reliability of the array

Trade Off Table

This trade-off table evaluates three innovative solutions, providing a concise comparison to guide viewers in determining prototype practicality and efficiency for different environmental and operational contexts.

Table 3: Trade Off Table

System	Scalability	Portability
Origami Solar Array	High	High
Compact Deployable Box	Low	Medium
Steamer-Inspired Structure	Medium	High

Conclusion

The project successfully demonstrated that the origami-inspired solar array system significantly enhances portability, deployment efficiency, and space utilization compared to conventional box-type designs. The transportable system achieved improvements in power-to-weight ratios and reliable deployment cycles – crucial for long-term space missions. The integration of an origami-based design, coupled with the strategic use of lightweight materials, enabled a compact stowage solution and efficient deployment of the system. Looking ahead, future research should focus on advanced material testing, further refinement of the deployment mechanism, and the integration of other spacecraft systems to scale the design for larger, more demanding deployments in future Artemis missions.

Acknowledgement

The Solar Fold Team would like to thank our faculty advisor, Professor Erik Siggelkoe, and lab technician, Oscar Pena. We would like to thank the NASA MUREP Program. Specifically, we would like to acknowledge the funding from NASA MUREP Innovative New Designs for Space (MINDS) Activity.